# Vericoding

## Verification-Driven Development

From behavioural features to machine-checked proofs

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## Vibecoding

The status quo of LLM-driven development.

**You prompt. The model emits. It runs.**

What you leave behind:

- **No specification** of intended behaviour
- **No evidence** of correctness
- **No conformance regime** for the next generation

The only durable artifact is the code itself.

## Vericoding

The methodology specsaver operationalises.

**You write a specification. The spec survives.**

What you leave behind:

- **Features** — Gherkin scenarios domain experts can review
- **Contracts** — mathematical pre/post-conditions
- **Proof obligations** — machine-checked in Rocq/Coq

Generated code is a replaceable implementation detail.

# The Pipeline

1. **Feature** — Gherkin scenario tables define what must work
2. **Implementation** — write the production code, SQLAlchemy / logging / OTel
3. **Testing (sanity)** — make sure it basically works
4. **Contract** — capture observed behaviour as mathematical spec
5. **Testing (dialectic)** — re-run under contract checking;
   either the contract or the implementation is wrong → reconcile
6. **Formal lockdown** — lower to Coq, LLM closes obligations;
   DISPROVED yields witnesses → back to step 5

# Two Development Flows

## Retrospective (typical)

- **Feature** — Gherkin scenario tables
- **Implementation** — write the production code
- **Testing (sanity)** — make sure it basically works
- **Contract** — capture observed behaviour as mathematical spec
- **Testing (dialectic)** — re-run under contract checking; either contract or implementation is wrong — the dialectic
- **Formal Proof** — lower to Coq, LLM closes obligations; DISPROVED → back to dialectic

## Contract-First

- **Feature** — Gherkin scenario tables
- **Contract** — write spec as acceptance criteria
- **Implementation** — build code against the specification
- **Testing** — validate that implementation satisfies contract
- **Formal Proof** — machine-checked against kernel

Both converge: the contract is the lasting artifact.
DISPROVED yields witnesses → back to the dialectic.

# Contract Language

## The pure terminating fragment of Python

Every clause is a boolean-valued lambda:

- `requires(state, args)` — admissibility
- `ensures(state, args, result, new_state)` — success post
- `when(state, args)` — exception conditions
- `invariant(state)` — global legality

**Static purity check** rejects side effects, loops, and non-boolean returns before execution.

# Anatomised as mathematical record

$$C = \langle \Sigma,\ args,\ \textsf{pre},\ \textsf{post},\ \mathcal{X},\ \mathcal{I},\ \mathcal{D},\ \mathcal{W},\ \mathcal{R},\ \mathcal{G} \rangle$$

$\mathcal{X}$ — exceptional exits (when + ensures + frame)

$\mathcal{I}$ — invariants on every state

$\mathcal{D}$ — derived-state definitions

$\mathcal{W}$ / $\mathcal{R}$ — semantic frame

$\mathcal{G}$ — ghost state

# Semantic Frames

Declare **what you write**, not what you don't.

## Write paths

```
writes = {
    "state.products[sku].reserved",
    "state.invitations[token]",
    "state.reservation_log",
}
```

## Derived obligations

✓ Every other keyed attribute unchanged

✓ Every other keyed row untouched

✓ All other event channels silent

✓ Deletions always rejected

✓ Keyed-row inserts allowed for args-resolved keys

The checker derives frame soundness from write paths alone. No "everything else unchanged" clauses needed.

# Exception Exits

Exceptions are **data**, not control flow.

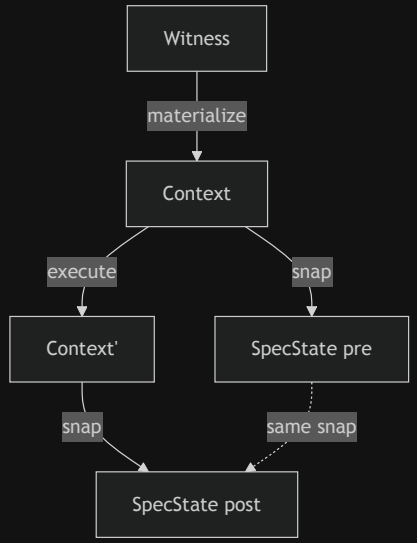
```
exceptions = [
    ExcExit(
        raises = InsufficientStockError,
        when   = [lambda s, a:
                    s.products[a.sku].on_hand
                  - s.products[a.sku].reserved < a.quantity],
        writes = {"state.failure_log"},
        ensures = [lambda s, a, e, s':
                    extends_by_one(
                       s.failure_log, s'.failure_log,
                       λf. f.sku == a.sku
                           ∧ f.reason == e.code)]
    )
]
```

The runner checks: correct exception raised, failure telemetry contains exactly one new event with exact fields, nothing outside the exit's writes changed.

# Symmetric Projection

**One function, used twice.**

The projection `snap(ctx) → SpecState` reads the concrete world (SQLite DB + event log) into an immutable spec state. It is called *identically* before and after execution — no separate "read" and "check" path.




Contracts compare pre and post produced by the same projection function. The service is ordinary SQLAlchemy code — it knows nothing about specs.

# Symmetric Projection — Runner

1. materialize

   witness → temp SQLite DB + engine + EventLog

2. pre-check

   invariants on pre-state; admissibility (requires)

3. execute

   service runs against real SQLite via SQLAlchemy;
   wrapper emits typed events into the log

4. frame check

   everything outside writes unchanged

5. derived check

   derived fields ≡ recomputed from observed

6. post-check

   ensures (or exit ensures) for the outcome;
   invariants on post-state

# Domain Declaration

One object → runners, CLI, conformance suite.

```
inventory = SqlDomain(
    name         = "inventory",
    package      = "examples.inventory",
    materializer = SqlMaterializer(ddl, TableSpec[]),
    projection   = SqlProjection(types, hooks),
    operations   = [
        SqlOperation(contract, wrapper,
                     witness_builder, feature_file),
        ...
    ]
)
```

scenario runners | CLI dispatch | conformance tests

Adding a domain: types + service + contracts + features + witness builders + one declaration. Everything else (materializer, projection, runner wiring, test dispatch) is derived.

# Theory Stack

Services run **unmodified** on differentially-tested stubs.

**Service** ordinary SQLAlchemy / logging / OTel API code

**Shim** make_engine, make_log_capture, make_otel_capture

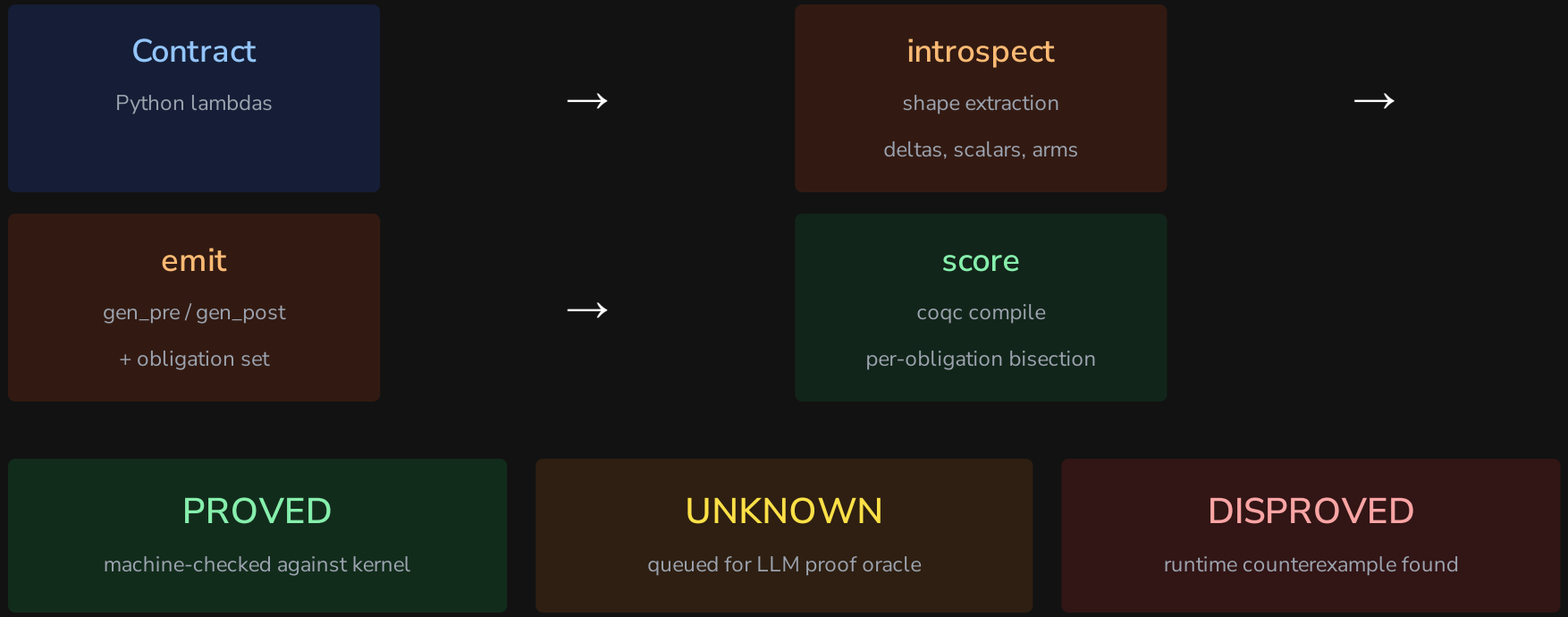
**StubHandler** operational semantics over the state model

**State model** TableStore / LogStore / OtelStore

Trace final interpretation (events)

**Theory conformance:** Stub and real library agree observationally on a differential suite; translator rejects out-of-fragment input loudly; state model and trace are pure data.

# Lowering Pipeline



# Example: Inventory Reserve

Pre-condition:

$$\mathsf{pre}(s, a) \;=\; a.quantity > 0 \;\wedge\; a.sku \in s.products$$

Post-condition (excerpt):

$$\begin{aligned}\mathsf{post}(s, a, r, s') \;&= s'.products[a.sku].reserved \;=\; s[\,\cdot\,].reserved + a.quantity \\ &\wedge\; \mathsf{extends\_by\_one}(s.reservation\_log,\; s'.reservation\_log,\; \lambda e.\; e.sku = a.sku) \\ &\wedge\; \mathsf{implies}(crossed(s, a, s'),\; \mathsf{extends\_by\_one}(s.alert\_log,\; s'.alert\_log,\; \ldots))\end{aligned}$$

Exception exit:

$$\mathcal{X} = \{\, \langle \mathsf{InsufficientStock},\; s.available(a.sku) < a.quantity,\; \{failure\_log\},\; \mathsf{failure\_ensures} \rangle \,\}$$

# Example: Invitations

**Invite** (insert frame):

$$\begin{aligned}\textsf{post}_{invite}(s, a, r, s') = {} & s'.invitations[a.token].status = \text{pending} \\ & \wedge\ s'.invitations[a.token].expires\_at = a.now + 7 \cdot 86400\end{aligned}$$

**Accept** (multi-table delta):

$$\begin{aligned}\textsf{post}_{accept}(s, a, r, s') = {} & s'.invitations[a.token].status = \text{accepted} \\ & \wedge\ s'.members[a.user\_id].org\_id = s.invitations[a.token].org\_id \\ & \wedge\ s'.users[a.user\_id].default\_org = s.invitations[a.token].org\_id\end{aligned}$$

Design decisions for verifiability: Token is a caller-supplied argument (args-resolved insert key). now is an integer epoch argument (expiry is a pure comparison).

# Verified State

| Domain | Tables | Ops | Rows | Obligations |
|---|---|---|---|---|
| inventory | 1 | 3 | 22 | 6/6, 6/6, 4/4 |
| bank_transfer | 2 | 1 | 11 | 7/7 |
| invitations | 3 | 2 | 9 | runtime-only |
| **Total** | | 6 | 42 | 23 in 23 |

All four store-obligation contracts fully proved in Rocq. Trace obligations (extends_by_one) active in development.

```
Definition gen_post (sigma : sn_state)
    (vs : list sn_val) (r : Result)
    (ups : cell_updates) : Prop :=
  exists sku order quantity store_d
         on_hand_sku reserved_sku
         reorder_point_sku,
    vs = [LitString sku;
          LitString order;
          LitInt quantity] ∧
    sigma !! store_loc =
      Some (LitDict store_d) ∧
    dict_lookup_str sku store_d =
      Some (row_of on_hand_sku
                   reserved_sku
                   reorder_point_sku) ∧
    r = RVal (LitInt reserved_sku) ∧
    ups = [(store_loc,
            LitDict
              (dict_insert_str sku
                (row_of on_hand_sku
                  (reserved_sku + quantity)
                  reorder_point_sku)
                store_d))].
```

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